

DIY Love Messengers: Interactive Bots to Playfully Enrich Human Connections

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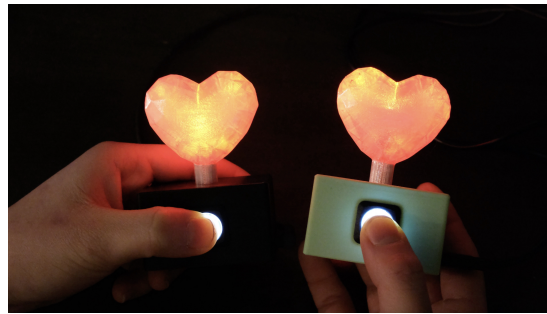


Figure 1: Two Love Messengers

ABSTRACT

Our pair of DIY Love Messengers are 3D-printed, small and light devices that allow users to communicate with each other, no matter how far apart they are around the world. The devices are powered by ESP32 microcontrollers, which allow direct communication between one another using Internet of Things (IoT) technology. The microcontrollers are able to send and read data from each other via a real-time database that stores the information sent by each device. They are also extremely straightforward in functionality - when either user presses the button on their Love Messenger, the lights on both Love Messengers light up simultaneously. This is a simple gesture that enriches relationships and connections, and is a meaningful gift for friends, loved ones and especially any form of long-distance relationship. The DIY Love Messengers are completely open-sourced and beginner-friendly to build, and are a simple application to demonstrate the interactive possibilities of IoT in enriching human relationships.

KEYWORDS

Interactive; Play; Internet of Things (IoT); Human Relationships

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1 INTRODUCTION

IoT refers to a network of physical devices or objects embedded with sensors, software, and network connectivity, enabling them to collect and share data. This paradigm shift has helped transform traditional lifestyles into more automated ones. For instance, the Smart Oven facilitates easy control of cooking functions, farmers utilize sensors to monitor crops, and governmental agencies track air pollution through data communication channels—all made possible by IoT. Its innovation has surged in popularity due to its automation, cost-efficiency, and convenient user interaction.

At the same time, IoT can foster playfulness. By streamlining data transfer and information exchange in the background, this technology can enhance human interaction and engagement. For example, our DIY homemade Love Messengers offer a simple yet impactful application—enabling users to send and receive small gestures of love daily through an LED lighting up. Due to its fully open-source nature, users can customize and modify these devices to incorporate additional functionalities and elements of play.

2 DESIGN AND IMPLEMENTATION

2.1 3D printing

Each Love Messenger consists of a transparent heart and an opaque encasing that are both 3D-printed. The LED sits in the transparent heart, while the ESP32 and other electronics reside in the box encasing.

2.2 Real-time database

In order for two or more ESP32s to communicate with each other from anywhere in the world, they need to be connected to the internet, and linked to a real-time database. This real-time database allows the ESP32s to store and read data sent and received from one another. We used Firebase as our real-time database, as it is

cloud-hosted and allows data to be stored as a JSON file, that we can then synchronize with the microcontrollers.

2.3 Assembly

We soldered the LED and button to each ESP32. Finally, we fit the LED into the 3D-printed transparent heart, and the button and ESP32 into the box encasing.

2.4 Coding the microcontrollers

We used the Arduino IDE and the Firebase Arduino Client Library for ESP8266 and ESP32 library to code the microcontrollers. The API key provided by Firebase enables the ESP32 to connect to our realtime database.

We coded the devices such that whenever a button on one device is pressed, its LED lights up and its ESP32 sends an integer to the database. The other ESP32 then instantaneously receives the data, and responds by having its LED light up. The devices can share and receive data as long as they are connected to a power supply and internet.

The full, complete documentation and files can be found in our GitHub repository. We also created a YouTube tutorial explaining the process of building these Love Messengers here.

3 DISCUSSION

Our DIY Love Messengers represent an affordable, enjoyable, and straightforward example of how IoT can enrich human relationships

and foster meaningful connections. However, it's crucial to address the security and privacy concerns associated with IoT technology. Notably, the platform we used to host our real-time database, Firebase, is owned by Google, raising potential concerns about data usage.

That said, our project operates on a small scale, and the information shared with the real-time database is not sensitive, mitigating significant privacy and security risks. We also aspire for hardware enthusiasts to leverage our documentation and tutorials to customize the Love Messengers according to their preferences. By making our project open-source, we aim to add to the community of enthusiasts who constantly share knowledge and collaborate. Our goal is to ensure that the joy and fun associated with our project remain accessible to everyone.

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